# A Hybrid Loop–Tree Duality / Capatti–Feynman–Forest Representation for Repeated Propagator Channels

Detailed derivation, tau-contour structure, dotted box example, and Python implementation status

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**Abstract**

This note records a self-contained derivation of the hybrid representation implemented in the current DOT-only prototype. The purpose is narrow: repeated propagator powers are represented by distinct graph edges with identical momentum routing and identical symbolic mass key. For one-loop repeated-channel graphs the representation combines ordinary LTD on the non-repeated part with a local CFF/Fourier treatment of the repeated channel. The central point is the order of operations. Localizing deltas are written in Fourier form. At fixed Fourier parameters, the energy integrals are ordinary Feynman energy integrals multiplied by complex exponentials; the sign of the corresponding Fourier combination fixes whether the contour must be closed above or below. After this LTD step, the remaining Fourier integrals are cone integrals of the CFF type, shifted by the already-chosen LTD/on-shell chain data. In multi-loop graphs, however, the repeated-sector Fourier sums generally couple in the contour-closure inequalities; the implementation therefore uses a coupled CFF-cone kernel for those cases rather than the invalid product of independent local cones. The final orientation label stores the edge-sign string together with either local chain signs or a coupled-cone marker, and the exported JSON stores the resulting numerator substitution map explicitly for every original edge.

# Contents

# Chapter 1

# Theory

## 1.1 Input graph, routing, and ordinary four-dimensional integral

Let

$$G = (V, E_{\text{int}}, E_{\text{ext}})$$

be a directed graph read from a DOT file. Internal edges are labelled by their loop-momentum decomposition and by an optional symbolic mass key. If the internal edge $e$ carries momentum

$$q_e(k,p) = \sum_{r=1}^{L} \rho_{er} k_r + \sum_{a=1}^{n_{\text{ext}}} \eta_{ea} p_a, \qquad \rho_{er}, \eta_{ea} \in \mathbb{Z}, \tag{1.1}$$

then its energy component is

$$q_e^0(k^0, p^0) = \rho_e \cdot k^0 + \eta_e \cdot p^0. \tag{1.2}$$

For fixed spatial loop momenta $\vec{k} = (\vec{k}_1, \ldots, \vec{k}_L)$ and fixed external four-momenta $p_a$, define the on-shell energy

$$\omega_e(\vec{k}, p, m) = \sqrt{\vec{q}_e^{\,2} + m_e^2 - i0}. \tag{1.3}$$

In the implementation this is displayed as OSE$[e]$, whereas external energy components are displayed as $\mathrm{E}[a] = p_a^0$. The Feynman denominator is

$$D_e(q_e) = (q_e^0)^2 - \omega_e^2 + \mathrm{i}0 = (q_e^0 - \omega_e + \mathrm{i}0)(q_e^0 + \omega_e - \mathrm{i}0). \tag{1.4}$$

The starting integral is

$$I_G = \int \prod_{r=1}^{L} \frac{\mathrm{d}^4 k_r}{(2\pi)^4} \, \frac{N_G(\{q_e(k,p)\}_{e\in E_{\text{int}}}, \{p_a\})}{\prod_{e\in E_{\text{int}}} D_e(q_e(k,p))}. \tag{1.5}$$

The numerator is assumed to satisfy the usual CFF condition: as a function of any internal edge energy used in the local CFF representation, its degree is at most one. The implementation does not symbolically use this polynomial condition; it is used mathematically to justify the CFF numerator replacement rule.

## 1.2 Repeated channels: graph convention and lifted numerator

Repeated propagators are not represented by a `pow` attribute. Instead one supplies distinct internal edges. A repeated channel is an equivalence class

$$R_g = \{r_{g,1}, \ldots, r_{g,\nu_g}\} \subset E_{\text{int}}, \qquad \nu_g \geq 2, \tag{1.6}$$

whose members have the same canonical momentum routing up to overall orientation and the same symbolic mass key. The canonical routing is denoted by

$$Q_g^0(k^0, p^0) = \rho_g \cdot k^0 + \eta_g \cdot p^0. \tag{1.7}$$

For every member $r_{g,a}$ there is an orientation sign $\epsilon_{g,a} = \pm 1$ such that

$$q^0_{r_{g,a}}(k^0, p^0) = \epsilon_{g,a} Q_g^0(k^0, p^0). \tag{1.8}$$

For notational clarity most formulas below use the canonical orientation, i.e. they absorb $\epsilon_{g,a}$ into the definition of the copy variable. The implementation keeps the original source-edge orientation sign and applies it when it builds per-edge substitution maps.

The numerator is lifted to a function of the copied edge energies,

$$N_G(k,p) \quad \longrightarrow \quad N_G^\sharp\big(\{q_s\}_{s\notin \cup_g R_g}, \{q_{r_{g,a}}\}_{g,a}, \{p_a\}\big), \tag{1.9}$$

with the defining property

$$N_G^\sharp\Big|_{q^0_{r_{g,1}}=\cdots=q^0_{r_{g,\nu_g}}=Q_g^0} = N_G. \tag{1.10}$$

In the Python interface this lifted numerator is what the user accesses through the `edges[i]` argument: every original internal edge receives its own four-vector, even if several edges represent the same repeated channel.

## 1.3 Exact localization identity

Introduce one independent energy variable $z_{g,a}$ for each repeated copy and one common channel energy

$$z_g = Q_g^0(k^0, p^0). \tag{1.11}$$

For a single repeated set $R_g$, the identity used under the energy integral is

$$\begin{aligned}
&\frac{N_G(\dots, z_g, \dots)}{\big[D_g(z_g)\big]^{\nu_g}} \\
&\quad = \int \prod_{a=1}^{\nu_g} \frac{\mathrm{d}z_{g,a}}{2\pi}\, N_G^\sharp(\dots, z_{g,1}, \dots, z_{g,\nu_g}, \dots) \prod_{a=1}^{\nu_g} \frac{1}{D_{r_{g,a}}(z_{g,a})} \prod_{a=1}^{\nu_g} 2\pi\delta(z_{g,a} - z_g).
\end{aligned} \tag{1.12}$$

Here all copies have the same spatial momentum and the same mass key, so $D_{r_{g,a}}$ differs only by its copy label. Applying (1.12) to all repeated sets gives the strict equality

$$\begin{aligned}
I_G = &\int \mathrm{d}\Phi_{3L}(\vec{k}) \int \prod_{r=1}^{L} \frac{\mathrm{d}k_r^0}{2\pi} \int \prod_g \prod_{a=1}^{\nu_g} \frac{\mathrm{d}z_{g,a}}{2\pi}\, N_G^\sharp(\mathcal{Q}) \\
&\times \prod_{s\in S} \frac{1}{D_s(q_s^0(k^0,p^0))} \prod_g \prod_{a=1}^{\nu_g} \frac{1}{D_{r_{g,a}}(z_{g,a})} \prod_g \prod_{a=1}^{\nu_g} 2\pi\delta\big(z_{g,a} - Q_g^0(k^0,p^0)\big),
\end{aligned} \tag{1.13}$$

where $S = E_{\mathrm{int}} \setminus \cup_g R_g$ and

$$\mathrm{d}\Phi_{3L}(\vec{k}) = \prod_{r=1}^{L} \frac{\mathrm{d}^3\vec{k}_r}{(2\pi)^3}. \tag{1.14}$$

## 1.4 Fourier representation of the localizing deltas

Use

$$2\pi\delta(z_{g,a}-z_g)=\int_{-\infty}^{+\infty}\mathrm{d}\tau_{g,a}\ \exp\{\mathrm{i}\tau_{g,a}(z_{g,a}-z_g)\}. \tag{1.15}$$

Then (1.13) becomes

$$\begin{aligned}I_G=&\int \mathrm{d}\Phi_{3L}(\vec{k})\int\prod_{g,a}\mathrm{d}\tau_{g,a}\int\prod_{r=1}^{L}\frac{\mathrm{d}k_r^0}{2\pi}\int\prod_{g,a}\frac{\mathrm{d}z_{g,a}}{2\pi}\ N_G^{\sharp}(\mathcal{Q})\\&\times\prod_{s\in S}\frac{1}{D_s(q_s^0(k^0,p^0))}\prod_{g,a}\frac{1}{D_{r_{g,a}}(z_{g,a})}\exp\left\{\mathrm{i}\sum_{g,a}\tau_{g,a}z_{g,a}-\mathrm{i}\sum_{g}T_g(\tau)Q_g^0(k^0,p^0)\right\},\end{aligned} \tag{1.16}$$

with

$$T_g(\tau)=\sum_{a=1}^{\nu_g}\tau_{g,a}. \tag{1.17}$$

This equation is the useful starting point for the hybrid derivation. The $z_{g,a}$ variables are copy energies and appear only in simple Feynman denominators and in exponentials. The original loop energies $k_r^0$ appear in the non-repeated denominators and in the phase

$$\exp\left[-\mathrm{i}\sum_g T_g(\tau)Q_g^0(k^0,p^0)\right]. \tag{1.18}$$

## 1.5 The fixed-$\tau$ energy integral and contour closure

At fixed real $\tau$, the $k^0$ integral in (1.16) is an ordinary Feynman energy integral multiplied by a plane wave. Choose a hybrid energy basis

$$u=(u_1,\ldots,u_L)=(Q_{g_1}^0,\ldots,Q_{g_m}^0,\ell_1^0,\ldots,\ell_{L-m}^0), \tag{1.19}$$

where the first $m$ coordinates are independent repeated-channel energies and the remaining coordinates complete an invertible linear basis. Let $J_u=|\det(\partial u/\partial k^0)|^{-1}$. Then

$$\begin{aligned}&\int\prod_{r=1}^{L}\frac{\mathrm{d}k_r^0}{2\pi}\prod_{s\in S}\frac{1}{D_s(q_s^0(k^0,p^0))}e^{-\mathrm{i}\sum_g T_gQ_g^0(k^0,p^0)}F(k^0)\\&=J_u\int\prod_{j=1}^{L}\frac{\mathrm{d}u_j}{2\pi}\prod_{s\in S}\frac{1}{D_s(q_s^0(u,p^0))}e^{-\mathrm{i}\sum_{g=1}^{m}T_gu_g}F(u).\end{aligned} \tag{1.20}$$

For $u_g=x+\mathrm{i}y$ the exponential $e^{-\mathrm{i}T_gu_g}$ has modulus $e^{T_gy}$. Therefore

$$T_g>0\Rightarrow u_g\text{ is closed below},\qquad T_g<0\Rightarrow u_g\text{ is closed above}. \tag{1.21}$$

This is the crucial point. The Fourier parameters do not merely multiply the result by harmless phases; they select the LTD contour closure in every repeated-channel direction.

Define the chain sign

$$\chi_g=\operatorname{sgn}T_g\in\{+1,-1\}. \tag{1.22}$$

For fixed $\chi=(\chi_1,$
$ldots,\chi_m)$, the $u$-integral is precisely an ordinary simple-pole LTD energy integral with contour closure vector

$$\kappa_g(\chi)=\begin{cases}\text{below}, & \chi_g=+1,\\ \text{above}, & \chi_g=-1,\end{cases} \tag{1.23}$$

for repeated-channel coordinates, and arbitrary fixed closure choices for the remaining basis coordinates. Hence the fixed-$\tau$ equality can be written as

$$\int \mathrm{d}^L k^0 \; e^{-\mathrm{i}T\cdot Q(k^0)}\, \mathcal{I}_{\text{simple}}(k^0) = \sum_{b\in\mathcal{B}(\chi)} \mathrm{LTD}_b^{\kappa(\chi)}[\mathcal{I}_{\text{simple}}] \; \exp\{-\mathrm{i}T\cdot Q_b^0(\chi)\}, \tag{1.24}$$

where $b$ runs over the ordinary LTD residue bases allowed by that closure vector, and $Q_b^0(\chi)$ denotes the repeated-channel energy evaluated on the residue solution of $b$. No higher-order pole appears in (1.24); all denominators are simple because the repeated propagators were replaced by independent copy variables.

## 1.6 Copy-energy residues and the origin of edge signs

For each copy variable $z_{g,a}$ one has the elementary contour integral

$$Z_{g,a}(\tau_{g,a}) = \int \frac{\mathrm{d}z}{2\pi} \frac{e^{\mathrm{i}\tau_{g,a}z}}{z^2-\omega_{g,a}^2+\mathrm{i}0} F(z). \tag{1.25}$$

The exponential $e^{\mathrm{i}\tau z}$ decays in the upper half-plane for $\tau > 0$ and in the lower half-plane for $\tau < 0$. With the usual Feynman prescription this gives

$$\tau > 0 \Rightarrow z = -\omega + \mathrm{i}0, \qquad \tau < 0 \Rightarrow z = +\omega - \mathrm{i}0. \tag{1.26}$$

It is convenient to introduce the edge orientation sign

$$\sigma_{g,a} = -\,\mathrm{sgn}(\tau_{g,a}), \tag{1.27}$$

so that the residue substitution is simply

$$z_{g,a} = \sigma_{g,a}\omega_{g,a}. \tag{1.28}$$

The residue also contributes the usual half-edge factor $(2\omega_{g,a})^{-1}$ and a sign determined by the chosen contour; this sign is part of the orientation prefactor stored in the JSON.

Now write

$$\tau_{g,a} = -\sigma_{g,a}\alpha_{g,a}, \qquad \alpha_{g,a} > 0. \tag{1.29}$$

Then

$$T_g(\tau) = -\sum_{a=1}^{\nu_g} \sigma_{g,a}\alpha_{g,a}. \tag{1.30}$$

Thus the chain sign is fixed by the sign of a single real linear form in the positive variables:

$$\chi_g = -\,\mathrm{sgn}\left(\sum_{a=1}^{\nu_g} \sigma_{g,a}\alpha_{g,a}\right). \tag{1.31}$$

If all copy signs $\sigma_{g,a}$ are equal, only one value of $\chi_g$ has a non-empty cone. If the signs are mixed, both $\chi_g = +1$ and $\chi_g = -1$ are possible. This proves the key compression used in the implementation: for each repeated set one needs only one additional binary chain sign, not one extra hidden sign per copy.

## 1.7 The tau cones and the local CFF factor

Insert (1.24) and (1.28) into (1.16). For each repeated set $g$, each copy-sign vector $\sigma_g = (\sigma_{g,1}, \ldots, \sigma_{g,\nu_g})$, and each chain sign $\chi_g$ define the cone

$$\mathcal{C}_g(\sigma_g, \chi_g) = \left\{ \alpha_g \in \mathbb{R}^{\nu_g}_{>0} \;\middle|\; -\operatorname{sgn}\left(\sum_a \sigma_{g,a}\alpha_{g,a}\right) = \chi_g \right\}. \tag{1.32}$$

The phase in this cone has the form

$$\begin{aligned}\Phi_{g,b}(\alpha_g; \sigma_g, \chi_g) &= \sum_{a=1}^{\nu_g} (-\sigma_{g,a}\alpha_{g,a})(\sigma_{g,a}\omega_{g,a}) - T_g(\tau)Q^0_{g,b}(\chi) \\ &= -\sum_a \alpha_{g,a}\omega_{g,a} + \left(\sum_a \sigma_{g,a}\alpha_{g,a}\right) Q^0_{g,b}(\chi) \\ &= -\sum_a \alpha_{g,a}\big(\omega_{g,a} - \sigma_{g,a}Q^0_{g,b}(\chi)\big).\end{aligned} \tag{1.33}$$

Therefore the remaining Fourier integral over the cone is

$$\mathrm{CFF}_g(\sigma_g, \chi_g; Q^0_{g,b}) = \int_{\mathcal{C}_g(\sigma_g,\chi_g)} \prod_{a=1}^{\nu_g} \mathrm{d}\alpha_{g,a} \ \exp\left[-\mathrm{i}\sum_a \alpha_{g,a}\big(\omega_{g,a} - \sigma_{g,a}Q^0_{g,b}(\chi)\big)\right]. \tag{1.34}$$

The integral (1.34) is exactly a CFF cone integral: the cone is polyhedral, its facets are linear, and its evaluation is a sum/product of inverse linear causal surfaces. The implementation stores this factorized result as a tree of surface ids. In particular, if the cone is simplicial with primitive generators $v_j$, then one term has the form

$$\prod_j \frac{\mathrm{i}}{v_j \cdot A_{g,b}(\sigma_g, \chi_g) - \mathrm{i}0}, \qquad A_{g,b,a} = \omega_{g,a} - \sigma_{g,a}Q^0_{g,b}(\chi), \tag{1.35}$$

while non-simplicial cones are decomposed into a sum of such products. This is the same operation performed by the ordinary CFF graph-contraction algorithm; the only difference is the shift $Q^0_{g,b}(\chi)$ inherited from the earlier LTD step.

The product over independent factors $\mathrm{CFF}_g$ is a one-loop or otherwise decoupled repeated-sector formula. In a general multi-loop graph a chosen energy-basis coordinate $u_j$ sees the Fourier coefficient

$$\Lambda_j(\tau) = \sum_g c_{gj}T_g(\tau), \tag{1.36}$$

where $Q^0_g = \sum_j c_{gj}u_j + \text{external shifts}$. The closure condition is then $\operatorname{sgn}\Lambda_j$, not the sign of a single $T_g$. Dependent or overlapping repeated channels therefore define one coupled polyhedral cone in all repeated-copy $\alpha$-variables. Treating that cone as a product of the individual $\mathcal{C}_g$ misses residues; this is the failure mode exposed by the sunrise and nested-kite examples. The code consequently evaluates such sectors with the coupled CFF-cone kernel.

## 1.8 Hybrid orientation and numerator substitution map

An exported hybrid orientation is

$$\mathfrak{o} = (\varsigma_e)_{e\in E_{\mathrm{int}}} \;\big|\; (\chi_g)_{g\in\mathcal{R}}, \tag{1.37}$$

where

$$\varsigma_e = \begin{cases} \sigma_{g,a}, & e = r_{g,a} \in R_g, \\ s_c, & e = c \text{ is an ordinary LTD cut edge in the residual basis,} \\ 0, & \text{otherwise.} \end{cases} \tag{1.38}$$

The chain signs after the vertical bar are ordered by the minimum edge id in each repeated set. For a given $\mathfrak{o}$, the numerator substitution map is stored explicitly. It is defined by the following equations:

$$q^0_{r_{g,a}}(\mathfrak{o}) = \sigma_{g,a}\omega_{r_{g,a}}, \qquad r_{g,a} \in R_g, \tag{1.39}$$

$$Q^0_g(k^0, p^0) = Q^0_{g,b}(\chi_g), \qquad \text{chain equation for group } g, \tag{1.40}$$

$$q^0_c(k^0, p^0) = s_c\omega_c, \qquad c \text{ ordinary cut edge in the residual LTD basis,} \tag{1.41}$$

$$q^0_s(\mathfrak{o}) = \rho_s \cdot k^0_b(\chi, s_c) + \eta_s \cdot p^0, \qquad s \text{ all remaining non-repeated edges.} \tag{1.42}$$

Equations (1.40) and (1.41) form a square linear system for the loop energies $k^0_b$. Equation (1.39) is an edge-level override for the copied repeated edges: the copied edge energies are no longer constrained to be equal inside the numerator. This is the CFF/TOPT numerator replacement rule in edge-energy form. It is why the JSON stores both loop-energy substitutions and edge-energy substitutions.

## 1.9 The master formula

Let $\mathcal{B}$ denote the set of residual simple-pole LTD bases compatible with the repeated-channel basis choice, and let $\mathcal{O}_g(\sigma_g, \chi_g)$ be the non-empty copy-sign/chain-sign cones. The chain of equalities from the original four-dimensional integral to the hybrid spatial integrand is

$$\begin{aligned} I_G &= \int \mathrm{d}\Phi_{3L}(\vec{k}) \int \mathrm{d}^L k^0 \, \mathcal{I}^{4D}_G(k^0, \vec{k}) \\ &= \int \mathrm{d}\Phi_{3L}(\vec{k}) \int \mathrm{d}^L k^0 \int \mathrm{d}z \int \mathrm{d}\tau \, \mathcal{I}^{4D}_{\text{split}}(k^0, z, \tau; \vec{k}) \\ &= \int \mathrm{d}\Phi_{3L}(\vec{k}) \sum_{b\in\mathcal{B}} \sum_{\sigma_g} \sum_{\chi_g : \mathcal{C}_g \neq \emptyset} \left[ N^\sharp_G\big(\mathcal{Q}_{b,\sigma,\chi}(\vec{k})\big) \; J_b \; H_b \prod_{h\in\mathcal{H}_b} \frac{1}{S^{\text{LTD}}_h(\vec{k})} \prod_{e\in\mathcal{E}_b} \frac{1}{S^{\text{LTD}}_e(\vec{k})} \prod_g \text{CFF}_g(\sigma_g, \chi_g; Q^0_{g,b}) \right] \\ &\equiv \int \mathrm{d}\Phi_{3L}(\vec{k}) \; I^{(3D)}_{G,\text{hyb}}(\vec{k}). \end{aligned} \tag{1.43}$$

Here:

- $\mathcal{I}^{4D}_{\text{split}}$ is the exactly localized integrand in (1.16);
- $J_b$ is the determinant/Jacobian and half-edge prefactor of the ordinary simple-pole LTD residue basis $b$;
- $H_b$ is the residue sign from the contour closure vector determined by (1.21);
- $S^{\text{LTD}}_h$ and $S^{\text{LTD}}_e$ are ordinary dual/LTD denominator surfaces of the residual simple graph after repeated-channel chain data have been fixed;
- $\text{CFF}_g$ is the local cone integral (1.34), evaluated by the standard CFF contraction algorithm;
- $\mathcal{Q}_{b,\sigma,\chi}$ is the explicit substitution map (1.39)–(1.42).

The classification of a displayed surface

$$S = \sum_e c_e\omega_e + \sum_a d_a p^0_a + c \tag{1.44}$$

defines an $E$-surface when all non-zero internal coefficients $c_e$ have the same sign, and an $H$-surface when both signs occur. This is the convention used by the pretty printer.

# Chapter 2

# The one-loop dotted box in detail

## 2.1 Routing and repeated split

Consider a one-loop box whose ordinary edges are $0, 1, 2$ and whose repeated channel is represented by three copies $3, 4, 5$. Choose the repeated channel energy as

$$z = q_3^0 = q_4^0 = q_5^0 \tag{2.1}$$

on the localized support, and write the ordinary edge energies as

$$q_i^0 = z + \Delta_i, \qquad i = 0, 1, 2, \tag{2.2}$$

where each $\Delta_i$ is a fixed external-energy shift. The original energy integral at fixed $\vec{\ell}$ is

$$I_\square^0(\vec{\ell}) = \int \frac{\mathrm{d}z}{2\pi} \frac{N(z+\Delta_0, z+\Delta_1, z+\Delta_2, z, z, z)}{\prod_{i=0}^{2} D_i(z+\Delta_i)\ D_3(z)D_4(z)D_5(z)}. \tag{2.3}$$

The localized copy form is

$$\begin{aligned} I_\square^0(\vec{\ell}) = \int \frac{\mathrm{d}z}{2\pi} \prod_{a=3}^{5} \frac{\mathrm{d}z_a}{2\pi} \int \prod_{a=3}^{5} \mathrm{d}\tau_a\ N^\sharp(z+\Delta_0, z+\Delta_1, z+\Delta_2, z_3, z_4, z_5) \\ \times \frac{\exp\{\mathrm{i}\tau_3(z_3 - z) + \mathrm{i}\tau_4(z_4 - z) + \mathrm{i}\tau_5(z_5 - z)\}}{\prod_{i=0}^{2} D_i(z+\Delta_i) D_3(z_3) D_4(z_4) D_5(z_5)}. \end{aligned} \tag{2.4}$$

Let

$$T = \tau_3 + \tau_4 + \tau_5. \tag{2.5}$$

Then the $z$ integral carries the exponential $e^{-\mathrm{i}Tz}$, so the contour closure is determined by $\operatorname{sgn} T$ exactly as in (1.21).

## 2.2 Copy residues and copy signs

For $a = 3, 4, 5$, define

$$\sigma_a = -\operatorname{sgn} \tau_a, \qquad \tau_a = -\sigma_a \alpha_a, \qquad \alpha_a > 0. \tag{2.6}$$

Then the copy-energy residues give

$$z_a = \sigma_a \omega_a, \qquad a = 3, 4, 5. \tag{2.7}$$

The remaining phase involving $z$ is governed by

$$T = -(\sigma_3\alpha_3 + \sigma_4\alpha_4 + \sigma_5\alpha_5), \tag{2.8}$$

and hence

$$\chi = -\operatorname{sgn}(\sigma_3\alpha_3 + \sigma_4\alpha_4 + \sigma_5\alpha_5). \tag{2.9}$$

For example, for the copy signs $(\sigma_3, \sigma_4, \sigma_5) = (+, -, -)$ both values of $\chi$ occur in different regions of the positive $(\alpha_3, \alpha_4, \alpha_5)$ octant. For $(+, +, +)$ only one value occurs.

## 2.3 The z-contour and the shifted skeleton surfaces

For fixed $\chi$, the $z$ contour closure is fixed. The ordinary denominators are

$$D_i(z + \Delta_i) = (z + \Delta_i - \omega_i + \mathrm{i}0)(z + \Delta_i + \omega_i - \mathrm{i}0). \tag{2.10}$$

In this one-loop example, after using the global residue theorem one may choose the chain-pole representation, in which the repeated channel fixes the chain value

$$z = z_\chi, \qquad z_\chi = \chi\, \omega_\star, \tag{2.11}$$

where $\omega_\star$ is the canonical on-shell energy of the repeated channel. The skeleton denominators then become the dual factors

$$\begin{aligned} D_i(z_\chi + \Delta_i) &= (z_\chi + \Delta_i - \omega_i)(z_\chi + \Delta_i + \omega_i) \\ &= S_{i,-}(\chi) S_{i,+}(\chi), \end{aligned} \tag{2.12}$$

with

$$S_{i,-}(\chi) = \chi\omega_\star - \omega_i + \Delta_i, \qquad S_{i,+}(\chi) = \chi\omega_\star + \omega_i + \Delta_i. \tag{2.13}$$

These are precisely the surfaces that appear as the LTD skeleton part of the hybrid expression. Depending on the signs of the internal-energy coefficients they are displayed as either $E$- or $H$-surfaces.

This chain-pole shortcut is not a valid general multi-loop rule. After copy localization, repeated denominators live in the copy variables $z_{g,a}$; a repeated-channel value $z = z_\chi$ is not automatically an available pole of every remaining energy integral. The safe statement is the fixed-$\tau$ closure rule above, followed either by the validated one-loop local cone or by the coupled cone construction.

## 2.4 The remaining tau integral for the box

Using (2.11), the cone phase for the repeated group is

$$\begin{aligned} \Phi_\Box(\alpha; \sigma, \chi) &= -\sum_{a=3}^{5} \alpha_a\omega_a + (\sigma_3\alpha_3 + \sigma_4\alpha_4 + \sigma_5\alpha_5) z_\chi \\ &= -\sum_{a=3}^{5} \alpha_a \left(\omega_a - \sigma_a\chi\omega_\star\right). \end{aligned} \tag{2.14}$$

Thus the local repeated block is

$$\mathrm{CFF}_{345}(\sigma_3, \sigma_4, \sigma_5|\chi) = \int_{\mathcal{C}(\sigma,\chi)} \mathrm{d}^3\alpha\ \exp\left[-\mathrm{i}\sum_{a=3}^{5} \alpha_a(\omega_a - \sigma_a\chi\omega_\star)\right]. \tag{2.15}$$

The CFF contraction of this cone produces a factorized tree of inverse linear surfaces. The important point is that all ordinary-edge shifts in (2.13) are already fixed before the cone integral is evaluated. Therefore the local CFF block is shifted by the chain sign and by the chosen skeleton chain value.

## 2.5 Final dotted-box hybrid expression

Combining the residue prefactors, skeleton surfaces, local cone factor and numerator substitution map gives

$$I^{(3D)}_{\square,\mathrm{hyb}}(\vec{\ell}) = \sum_{\sigma\in\{\pm1\}^3} \sum_{\chi:\mathcal{C}(\sigma,\chi)\neq\emptyset} \left[ \frac{\mathcal{P}(\sigma,\chi)}{\prod_{i=0}^{2} S_{i,-}(\chi) S_{i,+}(\chi)} \ \mathrm{CFF}_{345}(\sigma|\chi)\ N^{\sharp}_{\sigma,\chi} \right], \tag{2.16}$$

where

$$N^{\sharp}_{\sigma,\chi} = N^{\sharp}\big(z_\chi + \Delta_0, z_\chi + \Delta_1, z_\chi + \Delta_2, \sigma_3\omega_3, \sigma_4\omega_4, \sigma_5\omega_5\big). \tag{2.17}$$

This is exactly the structure exported by the Python build command for the box:

$$\text{orientation label } 000\sigma_3\sigma_4\sigma_5|\chi, \tag{2.18}$$

with zero signs on ordinary edges, individual signs on repeated edges, and one chain sign after the vertical bar. The expression (2.16) is pointwise equal to the split-mass simple-propagator LTD representation in the limit where the repeated masses are identified, and it is also pointwise equal to the pure CFF representation of the same localized graph.

# Chapter 3

# Python implementation and tests

## 3.1 DOT-only input

The current package is DOT-only. The graph file provides internal edge labels such as

```
v0 -> v1 [label="k1 - p1", mass="mA"];
```

If the `mass` attribute is absent, the edge is massless. If `mass="mA"` is present, the numerical value is supplied later through `-masses '"mA":  ...'`. Equal mass labels are used to decide which repeated channels can be merged into repeated sets. The `pow` attribute is not supported.

Repeated channels are represented by distinct graph edges, not by a fake powered edge. For example, a triple repeated box channel has three separate internal edges with the same canonical routing and mass key. This is necessary because the orientation and numerator substitution map are edge-wise.

## 3.2 JSON structure

The exported JSON is intentionally dependent on the DOT graph. It stores only the non-redundant expression data:

- the family, backend, loop and external names;
- a surface cache, where each surface is a compact linear expression in internal OSE[$e$] and external E[$a$];
- a list of orientations;
- for every orientation, a factorized tree of surface ids;
- for every orientation, explicit loop-energy and edge-energy substitution maps.

The evaluator reconstructs spatial momenta and masses from the DOT file and the numeric inputs, then evaluates the JSON tree directly. Mutating either a surface in the JSON or an edge substitution in the JSON changes the result in the regression tests; this guards against accidentally bypassing the exported structure.

## 3.3 Pretty output

The command

```
python hybrid3d.py build --family hybrid --dot examples/box_pow3.dot --pretty
```

prints:

- graph and repeated-channel summary;
- a surface cache with each surface classified as `e` or `h`;
- an orientation summary;
- optional details for a single orientation with `-show-details-for-orientation LABEL`.

For example,

```
python hybrid3d.py build --family hybrid --dot examples/box_pow3.dot \
  --pretty --show-details-for-orientation '000+--|+'
```

shows the substitution map corresponding to (2.17).

## 3.4 The diagnostic test command

The command

```
python hybrid3d.py test --dot examples/box_pow3.dot
```

performs a three-way diagnostic:

1. build/evaluate the pure CFF structure;
2. build/evaluate the hybrid structure;
3. compare against a split-mass LTD sequence in which the repeated mass labels are temporarily separated and then sent back together.

The split-mass LTD sequence is the most conservative numerical reference, because before the limit all propagators are simple and ordinary LTD applies without higher-order residues.

## 3.5 Current test coverage

The regression suite covers:

- validation of all shipped DOT examples except the intentionally invalid noisy graph;
- absence of doubled external-energy shifts in CFF surfaces;
- presence and evaluation of LTD skeleton surfaces inside the hybrid tree;
- JSON-driven mutation sensitivity of both surfaces and edge substitutions;
- family-distinct structures for repeated-channel graphs;
- box tests with dot-product numerators on ordinary and repeated edges;
- the proper iterated sandwiched-bubble example with numerator insertions;
- edge-by-edge dot-product numerator comparisons on every valid shipped example;
- a Mercedes repeated-channel example.

## 3.6 Relation between theory and code

The correspondence between equations and code is:

| Theory object | Implementation object |
|---|---|
| Internal on-shell energy $\omega_e$ | Display token `OSE[e]` |
| External energy $p_a^0$ | Display token `E[a]` |
| Repeated set $R_g$ | Canonical routing/mass equivalence class of internal edges |
| Copy sign $\sigma_{g,a}$ | Edge sign before the vertical bar in `orient_label` |
| Chain sign $\chi_g$ | Sign after the vertical bar in `orient_label` |
| Cone integral $\mathrm{CFF}_g$ | Factorized tree nodes attached to the orientation |
| Skeleton surfaces $S_{i,\pm}$ | Surface ids in the same evaluated tree |
| Numerator substitution $\mathcal{Q}_{b,\sigma,\chi}$ | Stored `loop_q0` and `edge_q0` maps |

# Bibliography


[1] S. Catani, T. Gleisberg, F. Krauss, G. Rodrigo and J.-C. Winter, and subsequent multiloop LTD developments; see the LTD derivation used in arXiv:1906.06138.

[2] Z. Capatti et al., Capatti–Feynman–Forest / causal representation derivations, arXiv:2211.09653.

[3] Pedagogical box derivation of the CFF representation, arXiv:2311.14374.

[4] The `cLTD` Mathematica implementation and the gammaLoop CFF generation code, https://github.com/alphal00p/cLTD and https://github.com/alphal00p/gammaloop.